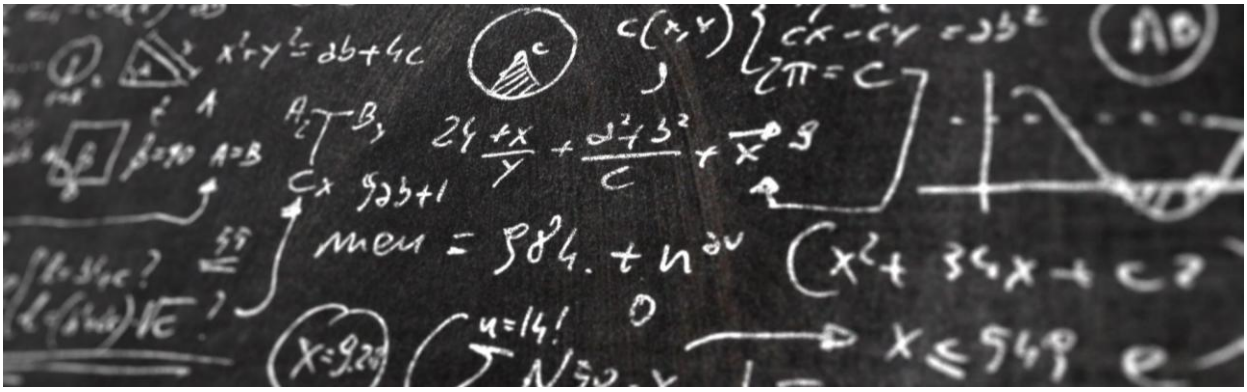


Welcome to 95219 - Writing Software For Machine Learning

Introduction:

This course is going to be a more technical course meant to gear you towards competency in developing a client/server software system. During the system's development, you will learn how to develop the code as a professional will.

The course will be composed of a hybrid of a lecture + practice session.

Each lecture is made up of one or more chapters.

Each chapter is made of a short video, lecture and hands-on. Some of the chapters will have mini-assignments

The course work will be done in teams of 3 students when submitting the assignments (Except for the first Mini HW, which will be submitted by each student individually).

In the course we will have **4** Mini HW, The Mini HW submissions should take you < 3 hours and come to help you practice the tools and methods learned in class in preparation for your HW and Project.

Next there will be one big HW that will serve as the foundation of the final project. After the HW you will submit a project proposal detailing the project you will be working on in the second half of the semester.

Perquisites for Week 1:

- Sign up to GitHub
- Download and set up WSL2 and make sure you can run Ubuntu on your laptop (You will need these for the first practice session.) See instructions in Moodle

Week Number	Topic	HW\Project	Mini HW
1	Git		Publish Mini HW1 – Git and Github
2	Containers		Submit Mini HW1 – Git and Github Publish Mini HW2 - Containers
3	Unit Tests		Submit Mini HW2 - Containers Publish Mini HW3 – Unit and Stress tests
4	Sockets + IPC		Publish Mini HW3 – Unit and Stress tests
5	Client Server + Python packages and organization	Publish HW	
6	More Python +native code vs. python		
7	Parallel Programming	Submission HW +Publish Project Prospal	Publish Parallel Programming
8	Debugging Tools + Methods	Submission Project Prospal	Submit Parallel Programming
9	Continuous integration (CI/CD)		
10	Linux Lab - 1		
11	Linux Lab - 2		
12	Final Project presentations		

Grading

Mini HW- 20%

HW – 30%

Final Project – 50%

Final Project Presentation

Each team is **required** to give an **8-minute** presentation for the app they are developing as part of the final project. This includes **5 minutes of presenting** and **3 minutes for Q&A.**

While the presentation itself is **mandatory**, it will not be graded directly. However, there's an **incentive:**

1. 1st place (as voted by the class) will earn +10 bonus points on the final project.
2. 2nd place will receive +5 bonus points.

After all presentations, the class will vote on the app they would be most likely to **"invest"** in.

Note: A live demo is optional, but highly recommend - it can significantly increase your chances of winning over your peers.

Course Staff:

Staff	email	Reception Hour	Zoom\Class
Noam Cohen	cnoam@technion.ac.il	Monday 11am	B 319
Elad	elad.kalif@campus.technion.co.il	Sunday 14:30-15:30	Class 152

The TA hours will be conducted via Zoom\Class depending on the student's requests, please email us before office hours so we know to be there.

If the time isn't comfortable for some students, please send an email and we will set another time to meet comfortable to you.